

UNIT – 3

OPPORTUNITY ASSESSMENT AND PROTOTYPE DEVELOPMENT

Identify and Map Global Competitors

Introduction

Global competitor mapping is the process of identifying and analyzing companies that compete in the same industry or market across different countries and regions. It helps businesses understand the competitive landscape, benchmark performance, identify opportunities, and develop effective strategies for growth and innovation.

Meaning of Global Competitors

Global competitors are companies that operate internationally and offer similar products or services to customers in multiple countries. These firms compete for market share, customers, resources, and brand recognition on a global scale.

Examples

- In smartphones: Apple Inc., Samsung Electronics, Xiaomi.
- In e-commerce: Amazon and Alibaba Group.

Steps to Identify Global Competitors

1. Define the Industry

- Determine the industry and market in which the business operates.
- Identify the products or services offered.

2. Identify Direct Competitors

- Companies offering similar products or services to the same customer segment.

3. Identify Indirect Competitors

- Companies offering alternative solutions to satisfy the same customer need.

4. Analyze Market Presence

- Examine competitors' geographical reach and international operations.

5. Study Customer Segments

- Identify the target markets and customer groups served by competitors.

6. Review Competitor Strategies

- Analyze pricing, marketing, distribution, innovation, and customer service strategies.

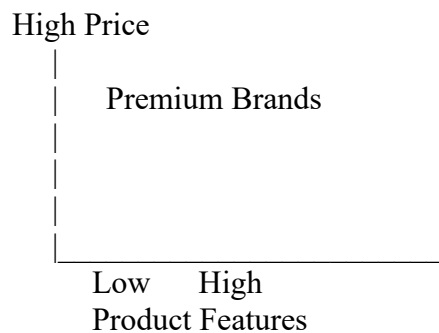
Competitor Mapping Framework

A. Market Share Mapping

Competitor Market Presence Market Share

Company A	Global	High
Company B	Regional	Medium
Company C	Local	Low

B. Product–Price Mapping



This helps identify where competitors are positioned in the market.

C. Strength–Weakness Mapping

Competitor	Strengths	Weaknesses
Competitor A	Strong brand image	High pricing
Competitor B	Low cost products	Limited innovation
Competitor C	Excellent customer service	Smaller market presence

Factors to Analyze While Mapping Competitors

1. Products and Services

- Quality
- Features
- Variety

2. Pricing Strategy

- Premium pricing
- Competitive pricing
- Discount pricing

3. Market Position

- Brand reputation
- Customer loyalty
- Market share

4. Distribution Network

- Online channels
- Retail stores
- International presence

5. Innovation Capability

- Research and development activities
- New product launches

6. Customer Experience

- Service quality
- Customer support
- User satisfaction

Benefits of Global Competitor Mapping

1. Better Strategic Planning

- Helps organizations formulate effective business strategies.

2. Identification of Market Opportunities

- Reveals underserved markets and customer needs.

3. Competitive Advantage

- Helps businesses differentiate themselves from competitors.

4. Benchmarking

- Enables comparison with industry leaders.

5. Risk Reduction

- Anticipates competitive threats and market changes.

Example: Online Education Industry

Global Competitors

- Coursera
- Udemy
- edX

Mapping Analysis

Company	Strength	Weakness
Coursera	University partnerships	Higher course costs
Udemy	Large course library	Variable course quality
edX	Academic credibility	Smaller course catalog

Opportunity

Provide affordable, industry-specific courses for emerging markets.

Conclusion

Identifying and mapping global competitors is an essential part of strategic management and entrepreneurship. It helps organizations understand the competitive environment, evaluate strengths and weaknesses, identify opportunities, and develop effective strategies. By continuously monitoring global competitors, businesses can improve their competitiveness, innovation, and long-term success in international markets.

Review Industry Trends and Understand Market Sizing: TAM, SAM, and SOM

Introduction

Before launching a new product, service, or business venture, entrepreneurs must analyze industry trends and estimate the potential market opportunity. Market sizing helps determine the scale of business potential and guides investment, marketing, and growth strategies. The most commonly used framework for market sizing is **TAM, SAM, and SOM**.

1. Review of Industry Trends

Meaning

Industry trends are the general directions in which an industry is moving due to changes in technology, customer preferences, economic conditions, regulations, and competition.

Importance of Reviewing Industry Trends

- Identifies emerging opportunities.
- Helps anticipate future market changes.
- Supports strategic decision-making.
- Reduces business risks.
- Encourages innovation and competitiveness.

Types of Industry Trends

A. Technological Trends

- Artificial Intelligence (AI)
- Automation
- Cloud Computing
- Internet of Things (IoT)

B. Economic Trends

- Inflation
- Consumer spending patterns
- Economic growth

C. Social Trends

- Digital learning
- Remote work culture
- Health and wellness awareness

D. Environmental Trends

- Sustainability
- Renewable energy
- Green manufacturing

E. Regulatory Trends

- Government policies
- Industry regulations
- Data privacy laws

Sources of Industry Trend Information

- Industry reports
- Government publications
- Market research studies
- Customer surveys
- Competitor analysis
- Trade journals and business magazines

2. Market Sizing

Meaning

Market sizing is the process of estimating the potential demand and revenue opportunity for a product or service within a specific market.

It helps answer:

- How big is the market?
- Who are the customers?
- What share of the market can the business realistically capture?

TAM, SAM, and SOM Framework

A. TAM (Total Addressable Market)

Meaning

TAM represents the total market demand for a product or service if the company could serve every potential customer globally.

Formula

TAM = Total Number of Potential Customers × Average Revenue per Customer

Example

Suppose there are 10 million students interested in online MBA courses worldwide.

Average annual course fee = ₹10,000

TAM = 10,000,000 × ₹10,000

= ₹100,000,000,000 (₹10,000 Crore)

Importance

- Shows the maximum possible market opportunity.
- Attracts investors by demonstrating market potential.

B. SAM (Serviceable Available Market)

Meaning

SAM is the portion of TAM that a company can realistically serve based on its business model, geography, and capabilities.

Example

If the startup focuses only on India and there are 2 million potential customers:

$$\text{SAM} = 2,000,000 \times ₹10,000$$

$$= ₹20,000,000,000 \text{ (₹2,000 Crore)}$$

Importance

- Reflects the actual target market.
- Helps define marketing and operational strategies.

C. SOM (Serviceable Obtainable Market)

Meaning

SOM is the portion of SAM that the company can realistically capture in the short term considering competition, resources, and market conditions.

Example

If the startup expects to capture 5% of the Indian market:

$$\text{SOM} = 5\% \text{ of } ₹2,000 \text{ Crore}$$

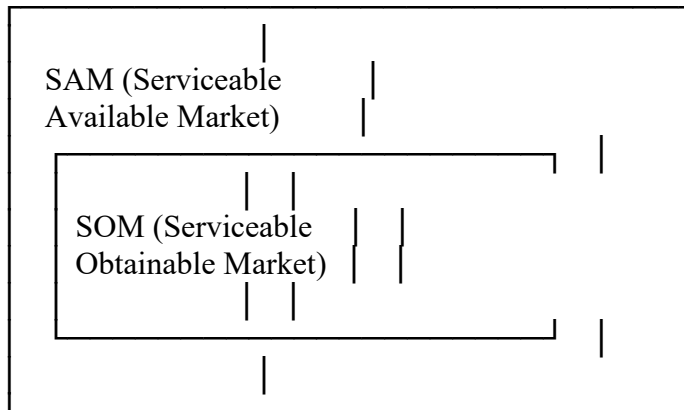
$$= ₹100 \text{ Crore}$$

Importance

- Provides realistic sales targets.
- Supports financial planning and forecasting.

Relationship Between TAM, SAM, and SOM

TAM (Total Addressable Market)



Example: Online Learning Startup

Business Idea

AI-based online learning platform for MBA students.

TAM

All MBA students globally interested in online education.

SAM

MBA students in India with internet access.

SOM

Students the startup can realistically acquire within the first 3–5 years.

Benefits of TAM, SAM, and SOM Analysis

1. Strategic Planning

- Helps define business goals and growth strategies.

2. Investor Attraction

- Demonstrates market potential and scalability.

3. Resource Allocation

- Supports efficient use of marketing and operational resources.

4. Revenue Forecasting

- Provides realistic sales and profit estimates.

5. Market Opportunity Assessment

- Helps evaluate whether a business idea is viable.

Challenges in Market Sizing

- Inaccurate market data.
- Rapidly changing customer preferences.
- Difficulty estimating future demand.
- Competitive uncertainties.
- Regional and industry variations.

Conclusion

Reviewing industry trends and conducting market sizing are critical steps in entrepreneurship and business planning. The **TAM, SAM, and SOM framework** helps entrepreneurs understand the total market opportunity, identify their target market, and estimate realistic market share. By combining trend analysis with market sizing, businesses can make informed decisions, reduce risks, and improve their chances of success in a competitive environment.

Assessing Scope and Potential Scale for the Opportunity

(Based on concepts commonly discussed in innovation and entrepreneurship literature by John R. Bessant, Ajay Batra, and Donald F. Kuratko.)

Introduction

After identifying and validating an entrepreneurial opportunity, the next step is to assess its **scope** and **potential scale**. This assessment helps entrepreneurs determine whether the opportunity can generate sustainable growth, profitability, and long-term value. It also assists in deciding whether the venture deserves further investment of time, effort, and resources.

Meaning of Scope of Opportunity

The **scope of an opportunity** refers to the extent and range of the market that can be served by a product, service, or business idea.

Scope Includes:

- Target customer segments.
- Geographic coverage.
- Product or service applications.

- Industry sectors served.
- Potential customer needs addressed.

Example

An online learning platform may initially serve MBA students but later expand to engineering students, professionals, and corporate training markets.

Meaning of Potential Scale

Potential scale refers to the ability of a business opportunity to grow and expand over time in terms of:

- Customers.
- Revenue.
- Market share.
- Geographic reach.
- Product offerings.

A scalable opportunity can increase revenue significantly without a proportional increase in costs.

Factors for Assessing Scope and Potential Scale

1. Market Size

The entrepreneur should evaluate:

- Total Addressable Market (TAM)
- Serviceable Available Market (SAM)
- Serviceable Obtainable Market (SOM)

A larger market generally offers greater growth opportunities.

Questions

- How many potential customers exist?
- Is the market growing?

2. Customer Need and Demand

The opportunity should solve a real and important customer problem.

Indicators

- Strong customer interest.

- Unmet market needs.
- Repeat purchase potential.

3. Industry Growth Trends

Analyze whether the industry is expanding or declining.

Growth Indicators

- Technological advancements.
- Increasing customer adoption.
- Rising industry revenues.

Example

Digital education and AI-based services are experiencing rapid growth.

4. Competitive Environment

Assess the level of competition.

Consider:

- Number of competitors.
- Market concentration.
- Entry barriers.
- Availability of substitutes.

Opportunities with moderate competition and clear differentiation often have greater potential.

5. Scalability of the Business Model

According to entrepreneurship scholars, scalable business models allow growth without proportionately increasing costs.

Characteristics

- Technology-enabled operations.
- Standardized processes.
- Digital delivery channels.
- Low marginal costs.

6. Availability of Resources

Evaluate access to:

- Financial resources.
- Human resources.
- Technology.
- Infrastructure.
- Strategic partnerships.

Insufficient resources may limit growth potential.

7. Profitability Potential

The opportunity should generate sustainable profits.

Assessment Areas

- Revenue streams.
- Cost structure.
- Expected profit margins.
- Break-even period.

8. Risk Assessment

Entrepreneurs should identify:

- Market risks.
- Financial risks.
- Technological risks.
- Regulatory risks.

Lower risk opportunities generally offer greater sustainability.

9. Innovation and Competitive Advantage

As emphasized in innovation literature, opportunities with unique value propositions have stronger growth prospects.

Sources of Advantage

- New technology.
- Superior customer experience.
- Cost leadership.
- Product differentiation.

Opportunity Assessment Framework

Assessment Factor	Key Question
Market Size	How large is the market?

Assessment Factor	Key Question
Customer Need	Is there a significant problem to solve?
Industry Growth	Is the industry expanding?
Competition	Can the business differentiate itself?
Scalability	Can the venture grow efficiently?
Resources	Are required resources available?
Profitability	Will the opportunity generate profits?
Risk	What challenges may arise?
Innovation	Is the solution unique and valuable?

Example

Opportunity

AI-Based Career Guidance Platform for Students

Scope

- School students
- College students
- Job seekers
- Educational institutions

Potential Scale

- National and international expansion
- Subscription-based revenue model
- Integration with online learning platforms
- Corporate partnerships

Assessment

- Large and growing market.
- Strong customer need.
- Scalable digital platform.
- High growth and profitability potential.

Benefits of Assessing Scope and Potential Scale

- Improves decision-making.
- Reduces entrepreneurial risk.
- Attracts investors and stakeholders.
- Supports strategic planning.
- Identifies growth opportunities.

- Enhances chances of long-term success.

Conclusion

Assessing the scope and potential scale of an opportunity is a critical step in entrepreneurship. Drawing from the principles discussed by John R. Bessant, Ajay Batra, and Donald F. Kuratko, entrepreneurs should evaluate market size, customer demand, industry trends, competition, scalability, resources, profitability, and risk before proceeding. A well-assessed opportunity provides a strong foundation for sustainable growth, innovation, and business success.

Understanding Prototyping and Minimum Viable Product (MVP)

(Based on concepts commonly discussed in innovation and entrepreneurship literature by John R. Bessant, Ajay Batra, and Donald F. Kuratko.)

Introduction

In entrepreneurship and innovation, transforming an idea into a successful product requires testing and validation before full-scale development. **Prototyping** and the **Minimum Viable Product (MVP)** are important tools that help entrepreneurs evaluate ideas, gather customer feedback, reduce risks, and improve products before investing substantial resources. These approaches promote learning through experimentation and support customer-centered innovation.

Prototyping

Meaning

A prototype is an early model, sample, or representation of a product, service, or process developed to test concepts, functionality, and customer acceptance before final production.

Prototyping allows entrepreneurs to visualize ideas and identify improvements at an early stage.

Objectives of Prototyping

- Test the feasibility of an idea.
- Understand customer reactions.
- Identify design flaws and technical issues.
- Improve product functionality.
- Reduce development costs and risks.
- Facilitate communication among stakeholders.

Types of Prototypes

1. Low-Fidelity Prototype

- Simple and inexpensive representation.
- Created using sketches, paper models, or storyboards.

Example: Drawing the screens of a mobile application on paper.

2. High-Fidelity Prototype

- Detailed and interactive model that closely resembles the final product.

Example: A working mobile application with basic features.

3. Physical Prototype

- Tangible model of a product.

Example: A 3D-printed product model.

4. Digital Prototype

- Software-based representation.

Example: Website or application mock-ups.

Benefits of Prototyping

- Encourages innovation and creativity.
- Identifies problems early.
- Improves product quality.
- Reduces uncertainty and risk.
- Saves time and resources.
- Enhances customer involvement.

Minimum Viable Product (MVP)

Meaning

A **Minimum Viable Product (MVP)** is the simplest version of a product that contains only the essential features necessary to solve a customer problem and obtain feedback from early users.

The objective is not to build a perfect product but to learn from customers as quickly and economically as possible.

Characteristics of an MVP

- Contains core features only.
 - Addresses a specific customer problem.
 - Requires minimal development effort.
 - Enables rapid customer feedback.
 - Supports continuous improvement.
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Objectives of MVP

- Validate business assumptions.
- Test customer demand.
- Reduce development costs.
- Accelerate market entry.
- Learn from real customer experiences.
- Improve product-market fit.

Steps in Developing an MVP

1. Identify the Problem

- Define the customer need or pain point.

2. Understand Customer Requirements

- Conduct market research and customer interviews.

3. Define Core Features

- Select only essential features needed to solve the problem.

4. Build the MVP

- Develop a basic functional version.

5. Launch and Test

- Introduce the MVP to target customers.

6. Gather Feedback

- Collect customer responses and suggestions.

7. Improve and Iterate

- Refine the product based on feedback.

Difference Between Prototype and MVP

Aspect	Prototype	MVP
Purpose	Test concepts and design	Test market demand and customer acceptance
Functionality	May not be fully functional	Functional with essential features
Users	Internal teams, stakeholders, test users	Real customers
Objective	Validate design and feasibility	Validate business model and customer demand
Stage	Early development	Pre-market or initial market stage

Example

Business Idea

Online Career Guidance Platform for Students

Prototype

- Screens showing registration, career assessment, and recommendations.
- Used to demonstrate the concept.

MVP

- Functional website with career assessment and recommendation features.
- Released to a small group of students for feedback.

Outcome

- Student feedback helps improve features and user experience before full-scale launch.

Importance in Entrepreneurship

According to innovation and entrepreneurship literature, prototyping and MVP development help entrepreneurs:

- Minimize uncertainty.
- Validate assumptions.
- Improve innovation success rates.
- Enhance customer-centric development.
- Reduce investment risk.
- Increase the likelihood of achieving product-market fit.

Challenges

- Selecting the right features for the MVP.
- Interpreting customer feedback correctly.
- Limited resources and time.
- Balancing speed and quality.
- Managing customer expectations.

Conclusion

Prototyping and Minimum Viable Products (MVPs) are essential tools in entrepreneurship and innovation. A prototype helps entrepreneurs visualize and test ideas, while an MVP enables them to validate customer demand and business assumptions with minimal resources. As emphasized by John R. Bessant, Ajay Batra, and Donald F. Kuratko, these approaches support experimentation, learning, and continuous improvement, helping entrepreneurs develop successful products and reduce the risks associated with new ventures.

Developing a Prototype: Testing and Validation

(Based on concepts commonly discussed in innovation and entrepreneurship literature by John R. Bessant, Ajay Batra, and Donald F. Kuratko.)

Introduction

In the innovation and entrepreneurship process, developing a prototype is a crucial step that transforms an idea into a tangible model for evaluation. Prototypes help entrepreneurs test assumptions, gather customer feedback, identify improvements, and validate whether the proposed solution effectively addresses customer needs. Testing and validation reduce uncertainty and improve the chances of creating a successful product or service.

Meaning of a Prototype

A prototype is an early version or model of a product, service, or process developed to demonstrate concepts, functionality, and design before full-scale production.

It serves as a learning tool that enables entrepreneurs to experiment with ideas and make improvements based on feedback.

Objectives of Prototype Development

- Visualize and communicate ideas.
- Test product functionality.
- Validate customer needs and assumptions.
- Identify technical and design issues.
- Reduce development risks and costs.
- Improve product quality before launch.

Stages of Prototype Development

1. Identify the Problem

Clearly define the customer problem or need that the product aims to address.

Example

Students require an affordable and flexible online learning platform.

2. Generate Solution Ideas

Develop possible solutions through brainstorming, design thinking, and market research.

Activities

- Brainstorming
- SCAMPER technique
- Customer interviews
- Competitor analysis

3. Design the Prototype

Create a preliminary model of the product.

Types

- Sketches
- Wireframes
- Mock-ups
- Physical models
- Digital models

Example

Designing wireframes for an online learning application.

4. Develop the Prototype

Build a basic version with key features.

Focus Areas

- Core functionality
- User interface
- Ease of use
- Customer requirements

Prototype Testing

Meaning

Prototype testing is the process of evaluating a prototype with users and stakeholders to determine whether it meets intended objectives and customer expectations.

Objectives of Testing

- Verify functionality.
- Identify usability issues.
- Gather customer feedback.
- Assess customer satisfaction.
- Evaluate technical feasibility.

Methods of Prototype Testing

1. User Testing

Potential customers interact with the prototype and provide feedback.

2. Focus Group Discussions

Groups of users discuss strengths and weaknesses of the prototype.

3. Observation

Observe users while they interact with the prototype.

4. Pilot Testing

Deploy the prototype in a limited environment before full launch.

5. Expert Review

Industry experts evaluate the design, functionality, and feasibility.

Prototype Validation

Meaning

Validation is the process of confirming that the prototype effectively solves the identified customer problem and meets market needs.

Validation answers the question:

"Are we building the right solution for the right customers?"

Steps in Validation

1. Collect Feedback

Gather information from users, customers, and stakeholders.

2. Compare Results with Objectives

Evaluate whether the prototype achieves its intended purpose.

3. Analyze Customer Responses

Identify common themes, suggestions, and concerns.

4. Test Business Assumptions

Validate assumptions regarding demand, usability, and value proposition.

5. Refine the Prototype

Modify and improve the design based on findings.

6. Repeat Testing

Conduct additional testing until customer needs are adequately addressed.

Validation Criteria

Criterion	Key Question
Problem Fit	Does the prototype solve a real customer problem?
Customer Acceptance	Do customers find value in the solution?
Usability	Is the product easy to use?
Feasibility	Can it be developed and delivered effectively?
Scalability	Can it support future growth?
Profitability	Does it have revenue potential?

Example

Business Idea

AI-Based Career Guidance Platform

Prototype

A simple web application that recommends career paths based on student interests.

Testing

- Conduct trials with college students.
- Observe user interactions.
- Gather feedback through surveys.

Validation Findings

- Students appreciate personalized recommendations.
- Users request additional career information and skill-development resources.

Improvement

Add career roadmaps and training recommendations.

Benefits of Prototype Testing and Validation

1. Reduces Risk

Identifies issues before large-scale investment.

2. Improves Product Quality

Enhances functionality and customer experience.

3. Increases Customer Satisfaction

Ensures solutions meet actual customer needs.

4. Supports Innovation

Encourages experimentation and learning.

5. Saves Time and Resources

Avoids costly mistakes in later stages.

Challenges

- Limited customer participation.
- Inaccurate or biased feedback.
- Resource constraints.

- Technical limitations.
- Difficulty interpreting customer responses.

Conclusion

Developing a prototype, testing it, and validating its effectiveness are essential steps in entrepreneurship and innovation. As highlighted in the works of John R. Bessant, Ajay Batra, and Donald F. Kuratko, these activities enable entrepreneurs to transform ideas into practical solutions, reduce uncertainty, and improve customer satisfaction. Through continuous testing, feedback collection, and refinement, entrepreneurs can develop products and services that successfully address market needs and achieve long-term business success.

Core Teaching Tools: Venture Activity, No-Code Innovation Tools, and Class Activity

Introduction

Entrepreneurship education focuses on experiential learning, innovation, and practical application of business concepts. Core teaching tools such as **Venture Activities**, **No-Code Innovation Tools**, and **Class Activities** help students develop entrepreneurial skills, creativity, problem-solving abilities, and business acumen. These tools bridge the gap between theoretical knowledge and real-world business practice.

1. Venture Activity

Meaning

A venture activity is a practical learning exercise in which students create, develop, and manage a business idea or startup project. It provides hands-on experience in entrepreneurship and business management.

Objectives

- Develop an entrepreneurial mindset.
- Apply business concepts in real-world situations.
- Enhance innovation and creativity.
- Improve decision-making and leadership skills.

Examples of Venture Activities

- Developing a business plan.
- Creating a startup prototype.
- Conducting market research.
- Organizing a campus business fair.
- Launching a student-led venture.
- Preparing investor pitches.

Benefits

- Provides practical business experience.
- Improves teamwork and leadership.
- Develops opportunity recognition skills.
- Enhances problem-solving and strategic thinking.
- Builds confidence in entrepreneurial decision-making.

2. No-Code Innovation Tools

Meaning

No-code innovation tools are software platforms that allow users to create websites, mobile applications, workflows, databases, and business solutions without traditional programming knowledge.

These tools enable entrepreneurs to rapidly build and test ideas with minimal technical expertise.

Common Applications

- Website development.
- Mobile app creation.
- Workflow automation.
- Database management.
- Prototype development.
- Customer relationship management (CRM).

Examples of No-Code Tools

- Bubble
- Glide
- Airtable
- Zapier
- Webflow

Benefits

Faster Development

- Build prototypes and applications quickly.

Cost Efficiency

- Reduces software development costs.

Accessibility

- Enables non-technical entrepreneurs to innovate.

Rapid Testing

- Facilitates MVP development and market validation.

Flexibility

- Allows easy modification and improvement.

3. Class Activity

Meaning

Class activities are structured learning exercises conducted within the classroom to encourage participation, collaboration, and application of entrepreneurial concepts.

Objectives

- Reinforce theoretical concepts.
- Promote active learning.
- Encourage teamwork and communication.
- Develop critical thinking skills.

Examples

Brainstorming Sessions

Generating innovative business ideas.

Case Study Analysis

Examining real-world business situations.

Role Plays

Simulating entrepreneurial scenarios and negotiations.

Group Discussions

Discussing market trends and business opportunities.

Business Model Canvas Exercises

Designing and evaluating business models.

Pitch Presentations

Presenting startup ideas to peers and faculty.

Benefits

- Enhances engagement and participation.
- Develops communication skills.
- Encourages collaborative learning.
- Strengthens analytical and problem-solving abilities.

Comparison of Core Teaching Tools

Teaching Tool	Main Focus	Learning Outcome
Venture Activity	Real business creation and management	Entrepreneurial skills and practical experience
No-Code Innovation Tools	Technology-enabled innovation	Rapid prototyping and digital entrepreneurship
Class Activity	Interactive classroom learning	Knowledge application and teamwork

Importance in Entrepreneurship Education

- Encourages experiential learning.
- Promotes innovation and creativity.
- Develops leadership and decision-making skills.
- Enhances problem-solving abilities.
- Supports startup creation and business development.
- Prepares students for real-world entrepreneurial challenges.

Conclusion

Venture activities, no-code innovation tools, and class activities are essential teaching tools in entrepreneurship education. Venture activities provide hands-on business experience, no-code tools enable rapid innovation and prototype development, and class activities strengthen conceptual understanding through participation and collaboration. Together, these tools help students develop entrepreneurial competencies, foster innovation, and prepare them for successful careers as entrepreneurs and business leaders.